

Fascial stretching

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Introduction

Stretching is but one of multiple activities that athletes, fitness enthusiasts and movers of all persuasions may engage in before, during and after training or event participation. It may be performed by a person alone, with partners, with a team, or performed in an assisted fashion with a trainer or a therapist. There are several concepts and methods of stretching, from passive, static stretching of isolated muscles to dynamic or ballistic stretching during multi-joint movements to name just a few. Research in the last 2 decades has shown that most stretching methods are able to improve flexibility and joint ROM (Harvey et al., 2002; Decoster et al., 2005; Thomas et al., 2018). However, static stretching of longer durations seems to hamper neuromuscular performance, while dynamic and active stretching methods improve performance (Behm and Chaouachi, 2011; Kay and Blazevich, 2012; Opplert and Babault, 2018).

In this chapter, a specific and comprehensive approach called fascial stretching is introduced and described. When integrated into a complex training regimen, such a method may assist in improving movement ability and athleticism while potentially speeding up recovery and reducing risks of injury (Schleip and Müller, 2013). Applied together with manual and movement therapy for rehabilitation after injury or surgery, we believe that fascial stretching can be used

for relief from common subjective complaints of pain, soreness, stiffness and tightness.

Stretching and the fascial system

Flexibility has long been seen as a matter of muscle extensibility only. However, our modern understanding of flexibility now also includes neurophysiological factors (Weppeler and Magnusson, 2010), as well as other non-muscular components such as peripheral nerves and the fascial system (Wilke et al., 2016; Nordez et al., 2017). A fascial approach to stretching uses the same principles and philosophy that all types of fascia-oriented training are based on. It is well-established that functional outcomes in movement, sport and fitness are greatly improved when incorporating whole body participation, in contrast to an isolated focus on single muscles and body parts.

Fascial stretching has been described as one of the four essential elements necessary to help build, repair and maintain connective tissue health and fitness for optimal movement (Schleip and Müller, 2013). Stretching can provide the necessary stimulus to affect tissue remodeling in the connective tissue (Benhardt and Cosgriff-Hernandez, 2009), entailing improved mechanical properties of fascia. This includes helping muscles to efficiently contract, lengthen and expand. Dynamic forms of stretching can both mentally prepare and physically “warm-up” the

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entire neuromyofascial system for imminent activity. Traditionally, static stretching has been used to aid cool down, repair stress, strain and injury from activity and restore normal function, although many of its claimed effects remain under question (Herbert and Gabriel, 2002; McHugh and Cosgrave, 2010; Herbert et al., 2011).

Stretching based on function

As pointed out, more research on applied stretching is needed to showcase the broad and deep spectrum of possibilities in prescribing personalized parameters for individual athletes and patients. This challenge is multiplied when considering how to apply stretching parameters to the broad spectrum of flexibility variations commonly found among groups or teams.

Therefore, function-based stretching depending solely on task participation lacks an important component unless a person's mobility type is also considered. In addition to several other factors like anthropometrics or genetic muscle fiber composition, the degree of mobility also often determines what sport or fitness activity an athlete engages in. For example, we will see lean people participate in rock climbing and solid people engage in power lifting.

If we use the extreme ends of the “flexibility in humans” spectrum as examples of how to apply stretching to improve connective tissue fitness and health, we come up with what have been described as the Viking and the Dancer types (Schleip and Bayer, 2017). Briefly, the Viking can be described as the least mobile but strongest type of person. These large, wide people are built more for stability than mobility and participate in sports and activities where their size and strength are an asset (e.g. weightlifting and athletic field sports such as the shot put and hammer).

The other extreme is the Dancer, who is extremely flexible and supremely mobile. Naturally, the approach which either one of these extreme types takes when stretching must be different, based on whether they (1) tend to be less or more mobile and (2) on their function. Other important considerations are that the Viking must not have increased flexibility at the expense of their functional stability otherwise they may lose power and strength. Likewise, the Dancer must not gain joint stiffness or myofascial tightness that decreases their functional flexibility to perform aerial splits.

While both extremes of the flexibility spectrum—the Viking and the Dancer—are commonly seen, there are many more people who fall somewhere in between these two extremes. Some people may be half Viking and half Dancer, with some individuals being one or the other in the top half as opposed to the bottom half of the same body!

The stretch spectrum

The variety of the different methods, techniques, parameters and applications of stretching is vast. For the limits of this chapter we will discuss active stretching, used for warm-up before activity and for post-activity recovery.

Activity preparation (warm-up)

Depending on the individual needs and the desired sport, it is generally advised that a warm-up consists of a period of aerobic exercise, followed by dynamic stretching, and ending with a period of movement similar to the upcoming activity (McGowan et al., 2015).

The National Strength and Conditioning Association describes two basic phases of the warm-up (Sands et al., 2012). The first is a gen-

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eral warm-up period. In sport this may consist of five minutes of slow aerobic activity such as jogging, skipping, or cycling. The second phase, activating and mobilizing, is analogous to the stretching component of a typical warm-up. The focus on mobility, or actively moving through a range of motion, requires a combination of motor control, stability, and flexibility and more closely relates to the movement requirements an athlete will face. Typically, this phase lasts between ten and twenty minutes, with the shorter time periods being more common (Sands et al., 2012).

Activity recovery

The goal of post-activity recovery is to return the athlete back to a pre-activity level of optimal

performance. A plethora of strategies are used in everyday practice, including sleep and nutritional strategies as well as active recovery such as light aerobic movement or stretching (Nédélec et al., 2013; Calleja-González et al., 2016). Although scientific data suggests that there is no effect for stretching to reduce muscle soreness (Herbert et al., 2011), it is still widely used as a recovery tool after exercise.

The Great 8 Stretch to Win® program

The Great 8 stretches (Figure 28.1) are one example of a global core mobility program of fascial stretches that integrate well with core dynamic stability programs commonly performed by athletes and fitness enthusiasts before activity participation.

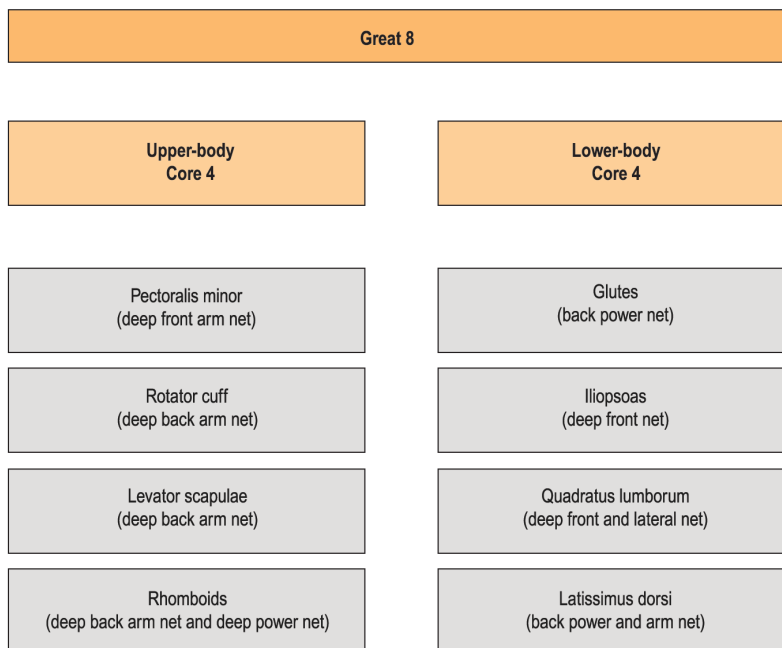


Figure 28.1

Structure and components of the Great 8

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We believe that achieving balanced mobility in the core regions of the body before training for stability will provide results that are more effective. This is because the program also serves as a self-assessment for the athlete. It empowers them to self-correct any imbalances that are discovered, which hopefully results in a decreased risk of micro-trauma, such as localized strain or stress accumulation.

The program is divided into the Core 4 of the lower body and the Core 4 of the upper body and target tissues lying within and around long chains or nets of fascia (see Figure 28.1). In contrast to other established approaches of myofascial chains we chose to use the term “fascial nets” (Frederick and Frederick, 2017).

For instance, the back power net (posterior diagonal line) includes the connection between the gluteal muscles and the contralateral lats.

While each stretch has the intent to mobilize and stretch a specific fascial net, it is also associated with a key local myofascial unit that (in our experience with athletes) is most directly responsible to improve functional movement.

Our focus for this chapter is on the Core 4 lower body part of the Great 8, because many sports and fitness activities engage these fascial nets to generate the power required for optimal performance.

What we call the Core 4 of the lower body anatomically includes the lumbar-pelvic-hip area. In terms of muscles, this would be all the core muscles and fasciae in the region of the lumbodorsal or thoracolumbar fascia, transversus abdominis, obliques, deep and superficial back extensors, iliopsoas, glutes and deep hip rotators. Our focus is on four key myofascial regions, which aim to improve mobility in professional athletes:

- Iliopsoas in the deep front net
- Glutes in the back power net
- Quadratus lumborum (QL) in the deep front, lateral and spiral net
- Latissimus dorsi in the back power net.

Lateral and back power nets: glute complex (Figure 28.2)

1. Sit on the floor and bend both knees, placing one leg in front and one behind.
2. Bring the front foot inward until the foot touches the back knee if possible.
3. Position your weight so you are sitting more on the glute of the front leg. Make any adjustments for comfort.
4. Place your hands in front of you in a push-up position with the arms straight to begin.
5. Lengthen your whole spine out from the top of your head as you inhale. Exhale moving down and forward over the knee into the stretch, keeping your spine long.
6. Flex, rolling up through spine to the upright start position. Repeat, taking the torso



Figure 28.2

Glute complex within back power net.

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forward to the left and right of knee at different angles targeting the different glute fibers.

7. Tip: Breathe, moving in and out of the stretch until you feel your tissues release. Drop your body down closer to the floor and move from side to side.

Spiral and lateral net: QL (Figure 28.3)

1. From the glute stretch, start walking your hands back until you feel a slight stretch in your back, hips or legs.
2. Keeping your hands still, lean to the front hand and inhale.
3. Exhale as you lean into the back hand, slightly bending the elbow.
4. Tip: Walk your hands slightly further away with each repetition to progress the stretch.

Deep front net: hip flexors (Figure 28.4)

1. From your last position, place your forearm back on the ground and find a stable position to balance, so there is full weight on that arm.



Figure 28.3

Quadratus lumborum within spiral and lateral nets.

2. Slide your forearm to the rear as your back starts to arch and stop when a mild stretch is felt.
3. Slightly leaning forward on both hands (not shown), inhale.
4. Exhale while arching back, looking up to ceiling. Repeat.
5. Tip: Lean further back to progress the stretch. Turn your chest towards the floor then to ceiling to stretch different angles. Find the stretch by arching your back, not by twisting.

Back power net: Latissimus dorsi (Figure 28.5)

1. Moving from the last position, inhale and then reach your arm up overhead.
2. Extend the arm out from the hip as you reach up and overhead, as though you are swimming in the air.
3. Exhale as you rotate your chest towards the floor, reaching your arm out.
4. Circle your arm down and back up overhead to repeat.



Figure 28.4

Hip flexors within the deep front net.

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Figure 28.5

Latissimus dorsi within back power net.

5. Tip: Keep reaching your arm throughout the stretch for maximal effect. Try to get your chest increasingly parallel to floor with each rep.

Fascial stretch technique

The difference in technique between traditional and isolated muscle stretching and that of fascial stretching can be summed up in a term we named The Stretch Wave™, which incorporates the following principles:

- Initiate and terminate body movement with eye movement
- Exhale on the stretch, inhale on the recoil
- Lead the stretch by breath volume and duration, not counting
- Perform tempo of stretch movement that matches the phase of training
- Undulate through the fascial net
- Oscillate across adjacent fascial nets
- Train elastic recoil by extending the fascial net movement intent.

Initiating and terminating body movement with eye movement may encourage full

neuromyofascial engagement from the mechanoreceptors in the skeletal eye muscles. They are connected to the deep neck muscles via the galea capitis and the superficial fascia of the neck (Stecco and Hammer, 2014). Potentially, this fascial connection can mediate an effect of eye movement on the neck and shoulder muscles, and encourages incorporating them into a holistic stretching and training program.

Exhale on the stretch, inhale on the recoil coordinates proper breathing with movement versus movement preparation.

Lead stretch by breath volume and duration, not counting encourages the athlete to experience a deeper, more interoceptive sense of timing. It uses the breath with the movement rather than just counting seconds. So, when the athlete performs slower breathing to match slower stretch movements, it may help shift the nervous system to the parasympathetic for better recovery. By contrast, faster, more dynamic movements accompanied by faster breathing better prepares the athlete sympathetically for imminent activity.

Perform tempo of stretch movement that matches phase of training parallels the description above in Lead stretch by breath volume and duration, not counting. In mimicking the movement velocity and amplitude according to the desired movement, the stretch serves as an optimal warm-up to enhance performance.

Undulate through the fascial net ensures full fascial engagement along the part of the net that is moving as well as being transmitted beyond.

Oscillate across adjacent fascial nets ensures that adjacent nets along septa and other connections between myofasciae are included for a full fascial net experience.

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Train elastic recoil by extending the fascial net movement intent encourages the athlete to maximize the extent of the specific movement to help train fascia's efficiency at storing potential energy and releasing kinetic energy in movement. This is carried out within the parameters specific to the activity so that any unnecessary stretching or overstretching is avoided. It also incorporates all of the previous techniques.

Below follows an example of how you would use the principles of fascial stretching described above. It is particularly effective at improving or maintaining maximum functional flexibility potential. Focus moving one key myofascial region at one attachment, release it then focus on moving the other attachment. Finish by combining both attachments for a full fascial net stretch, until there is no more gain in range of motion for that movement. Then move on to adjacent nets as indicated by your coach or practitioner, or based on your own self-assessment.

See the photographs and instructions below for an example of fascial stretching for the back net of the leg, targeting key muscles, hamstrings and gastrocnemius:

Proximal net focus (Figure 28.6)

- Use a bench, table or the floor, which facilitates a body position without tension or stretch in the target tissues. The foot is fixed on the chosen surface and the ankle is in a neutral position to start.
- Inhale to prepare then exhale as you perform a gentle anterior pelvic tilt until a mild stretch is felt at a proximal attachment



Figure 28.6

Proximal net stretch focus.

region. Avoid any other movement through the hip or spine. Inhale as you come out of the stretch then exhale as you repeat, until no further gain in movement is possible.

Distal net focus (Figure 28.7)

- Inhale to prepare then exhale and allow your pelvis to relax passively into a posterior pelvic tilt as you lean forward, flexing your spine and hip and lowering your head toward your knee until a mild stretch is felt. Inhale as you come out of the stretch then exhale as you repeat, until no further gain in movement is possible.

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Full net focus (Figure 28.8)

- Gently combine and simultaneously perform the previous proximal and distal stretches together. Inhale to come out of the stretch then exhale as you repeat, until no further gain in movement is possible.

Spread the stretch and add adjacent nets.

- Spread the stretch to access more of the target tissue: add options to include adjacent septa of connecting myofascial regions, by including gentle and small triplanar movements with circular undulations and/or oscillations.

- Add foot-ankle dorsiflexion and/or head-neck flexion one at a time or in combination for an even more comprehensive fascial stretch. (Note: that while this more extreme stretch may be appropriate and indicated with athletes like gymnasts, it will be contraindicated or unnecessary for most athletes because it is not part of their functional movement patterns. One example of a contraindication, among many, is a diagnosis of sciatica or any spinal disc lesion).

Repeat for all key muscle nets in the Core 4 or Great 8 program as needed or indicated.



Figure 28.7

Distal net stretch focus.



Figure 28.8

Full net stretch.

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Other important principles and concepts

- Experience a slow awareness of stretch without ever feeling pain. This will be more pronounced or apparent during slow tempo recovery phase stretching than pre-activity dynamic stretching.
- Creating a little extra space by tractioning your joints during recovery-phase stretching will increase the effectiveness and prolong the benefits of the stretch. Do this by gently lifting, reaching and/or expanding out before moving directly into a stretch.
- Do as many repetitions of a stretch as it takes before you no longer realize a gain in pre-activity or functional movement. Otherwise, try 3–6 progressive repetitions of each position. This can vary from side to side and from stretch to stretch.
- If there is any imbalance experienced like one-sided (unilateral) restriction to mobility, then it is advised that you correct it with more repetitions and/or duration of stretch movement until your reassessment shows the imbalance is resolved (see more details below).

How to correct functional imbalances in your fascial net

Stretching does not permanently alter muscle or connective tissue length. However, as

chronic stretching leads to increased stretch tolerance, functional flexibility asymmetries can be resolved. Often, you can easily and quickly correct imbalances discovered when you mobilize or stretch yourself. We have found that one of the easiest and quickest ways is to first test the stretch on each side to determine if one side is less mobile. Then start and end with the less mobile side, so that you get a 2-to-1 ratio. In other words, the tighter side gets a double dose. Over time, when you assess that both sides have the same mobility, you can go back to stretching at a 1-to-1 ratio for maintenance.

If the imbalance persists, increase the duration of the stretch. For instance, if you were using a slow stretch wave tempo then switch to a very slow stretch wave tempo so you spend more time releasing restrictions while you move through the stretch.

Also, keep in mind that with fascial stretching, sometimes the source of the problem is not locally, where you may feel pain or restriction, but is likely more distal or on the contralateral side.

A persistent imbalance also suggests that you add self-myofascial release like foam rolling, after you assess where you need to focus. This is described in more detail in Chapter 27. Table 28.1 lists self-stretch program parameters to help you design and integrate what you need for any phase of movement activity.

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Table 28.1 Self-stretch program parameters

Stretch program	Technique	Tempo	Intensity	Duration	Reps	Sets	Frequency	Notes
Preparation	Stretch Wave fast	Moderately fast	Light to moderate	Short exhale (1-2 counts)	3	1	Pre-training	- Adjust any parameter as needed. - Reps and sets are per side or for the whole body. - Adjust the program to what the body needs that day. - See Chapter 7 for self-programs and Chapter 8 for assisted programs.
Recovery and maintenance	Stretch Wave slow	Moderately slow	Moderate	Long exhale (3 counts)	3 or until there are no further gains in mobility	1	Pre-training	- Adjust any parameter as needed. - Reps and sets are per side or for the whole body. - Adjust the program to what the body needs that day. - See Chapter 6 for self-programs and Chapter 8 for assisted programs.
Restoration and correction	Stretch Wave very slow	Very slow	Moderate	Long exhale (3 counts)	3 or until there are no further gains in mobility	1	Every day until functional mobility goals are met	- Reps and sets are per body region. - Adjust the program to what the body needs that day. - If stretching increases pain, try stretching other regions of the same or nearby fascial nets, and then reassess. - If stretching increases tenderness, decrease all parameters and move more gently, and then reassess.

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